

Area of a circle



1

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|----------|--------------------------------|----------------------------------|----------|--|---------------------------------|
| a | i $64\pi \text{ cm}^2$ | ii 201.06 cm^2 | b | i $49\pi \text{ cm}^2$ | ii 153.94 cm^2 |
| c | i $25\pi \text{ cm}^2$ | ii 78.54 cm^2 | d | i $\frac{169\pi}{4} \text{ cm}^2$ | ii 132.73 cm^2 |
| e | i $324\pi \text{ cm}^2$ | ii 1017.88 cm^2 | f | i $16\pi \text{ cm}^2$ | ii 50.27 cm^2 |

2

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|-------------------------------|-------------------------------|--------------------------------|
| a $16\pi \text{ mm}^2$ | b $16\pi \text{ mm}^2$ | c $144\pi \text{ cm}^2$ |
|-------------------------------|-------------------------------|--------------------------------|

3

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|--------------------------------|--------------------------------|--------------------------------|
| a 254.47 cm^2 | b 452.39 cm^2 | c 452.39 cm^2 |
|--------------------------------|--------------------------------|--------------------------------|

4

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|----------|-------------------------------------|---|----------|--------------------------|---------------------------------|
| a | i 6 mm | ii $36\pi \text{ mm}^2$ | b | i 7 cm | ii $49\pi \text{ cm}^2$ |
| c | i $\frac{9}{\pi} \text{ cm}$ | ii $\frac{81}{\pi} \text{ cm}^2$ | d | i 11 mm | ii $121\pi \text{ cm}^2$ |

5

- | | | | |
|--|--|-----------------------------|-----------------------------|
| a $r = 8 \text{ cm}$ | b $r = \frac{6}{\sqrt{\pi}} \text{ cm}$ | c $r = 6 \text{ cm}$ | d $r = 4 \text{ cm}$ |
| e $r = \frac{5}{\sqrt{\pi}} \text{ mm}$ | | | |

6

- | | | | | | |
|----------|-------------------------|------------------------------|----------|--------------------------|------------------------------|
| a | i 5 cm | ii $10\pi \text{ cm}$ | b | i 11 cm | ii $22\pi \text{ cm}$ |
|----------|-------------------------|------------------------------|----------|--------------------------|------------------------------|

7

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|----------------------|----------------------------|
| a $r = 10.59$ | b 66.5 cm |
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8

- | | |
|---|----------------------------|
| a $\frac{12}{\sqrt{\pi}} \text{ mm}$ | b 6.77 mm |
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9

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|-------------------------|-------------------------|
| a 4 cm | b 8 cm |
|-------------------------|-------------------------|



10

a $\frac{5}{\sqrt{\pi}} \text{ mm}$

b $\frac{10}{\sqrt{\pi}} \text{ mm}$

11

a $\frac{24}{\sqrt{\pi}} \text{ mm}$

b 13.54 mm

12

a 14.1 cm²

b 19.6 mm²

c 95.0 m²

d 115.5 cm²

Applications

13 314.16 cm²

14

a 40 m

b 251.33 m

c 5026.55 m²

15

a 399 832.26 cm²

b 399 629.57 cm²

c Yes, because the approximate area is 202.69 cm² smaller than the actual area.